Amdahl's law

Little's law

Arithmetic Mean:

Harmonic Mean:

Geometric Mean:

non monotonic in property

$$GM = \sqrt[n]{x_1 \times \cdots \times x_n} = \left(\prod_{i=1}^{n} x_i\right)^{1/n} = e^{\left(\frac{1}{n}\sum_{i=1}^{n} \ln(x_i)\right)}$$

Use of a single number to characterize that system

SPEC Benchmarks:

Calculating Assessments → 3 steps

1| Compile system under test & run each program on it 3 times. Runtime measured & median value selected.

↓

avoid variation in execution time

2| Each 12 result normalize → calculating runtime ratio

$r_i = \frac{T_{refi}}{T_{suti}}$ → T_{refi} → reference run time

→ T_{suti} → system run time

3] $r_G = \left(\prod_{i=1}^{12} r_i \right)^{1/12}$ → geometric mean of the 12 run time.

→ overall metric.

Multi Processor System:

rate metric $\text{rate}_i = N \times \frac{T_{ref_i}}{T_{set_i}}$